ABERDARE TECHNICAL COLLEGE

LEVEL 5,MODULE 3

CAT2,TERM 3

ELEMENT 2:TEST COMPUTER SOFTWARE FUNCTIONALITY

UNIT OF COMPETENCE: INSTALL COMPUTER SOFTWARE

UNIT CODE: 0619 451 06A

**Total Marks:** 36 **Time:** 1 Hour 30 Minutes

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**Total Marks:** 40 **Time:** 1 Hour 30 Minutes

**QUESTION 1 (8 marks)**

**(a)** Define the term software testing. **(2 marks)**

**(b)** Define each of the following terms: **(3 marks)**

(i) Error

(ii) Bug (defect)

(iii) Failure

**(c)** State any three HTTP methods used in REST and give the purpose of each. **(3 marks)**

**QUESTION 2 (8 marks)**

**(a)** Explain the difference between verification and validation, giving one example of each. **(4 marks)**

**(b)** Using the restaurant analogy (customer, waiter, kitchen), explain what an API is and how works. **(4 marks)**

**QUESTION 3 (8 marks)**

**(a)** An age field on a registration form accepts values from 18 to 60 only. **(4 marks)**

(i) Use equivalence partitioning to identify the valid and invalid partitions. (2 marks)

(ii) Use boundary value analysis to list the six values you would test. (2 marks)

**(b)** A school website has the base URL https://school.com/api/. State the HTTP method and a suitable endpoint you would use in Postman to: **(4 marks)**

(i) Retrieve the list of students

(ii) Add a new student

(iii) Update a student's details

(iv) Delete a student record

**QUESTION 4 (8 marks)**

**(a)** Compare black box testing and white box testing under the following headings: knowledge of code, skills required, coverage, and what each type finds. **(4 marks)**

**(b)** A login form accepts passwords of 8 to 12 characters. Analyze how static testing and dynamic testing would each detect a missing check for the 12-character maximum, and explain why the two approaches complement each other. **(4 marks)**

**QUESTION 5 (4 marks)**

A bank is building a new system that must handle financial transactions with strong built-in security, while a sister company is building a public mobile app backend for simple data retrieval.

Evaluate which of REST or SOAP is better suited to each project. Justify your choices with at least two reasons per project. **(4 marks)**

1. (a) (i) evaluating a programme by running it to find it defects and check it meets requirements.

(ii) A mistake made by a developer

(iii) A design caused by an error

(iv) Its when the software behaves differently from what is expected while running.

(b) GET - retrieves data

POST - creates new data

DELETE - removes data

PUT - replaces data

2. (a) Verification checks that the product is built right against the specifications and without running it eg a code review

Validation checks that we are building the right product by running it against user needs eg user acceptance testing.

(b) API - is a go-between that tests two software system communicate. The customer is the client app, the waiter is the API and the kitchen is the server - The customer makes a request, a waiter carries it to the kitchen, the kitchen prepares it and the waiter returns the response - The customer never sees how the kitchen works.

3. (a) (i) Valid partition : 18-60 Invalid partition below 18 and above

(ii) Test values 17, 18, 19, 59, 60, 61

(b) (i) GET /api/students

(ii) POST /api/students

(iii) PUT /api/students/{id}

(iv) DELETE /api/student/{id}

4.